



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

A COMPREHENSIVE APPROACH TO MRI BRAIN IMAGE SEGMENTATION AND CLASSIFICATION USING DEEP LEARNING APPROACHES

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ABSTRACT

- Tumor grading and segmentation utilizing Magnetic Resonance Imaging (MRI) are necessary steps for correct diagnosis and treatment planning. There are many diversities in illumination across different Magnetic Resonance Imaging (MRI) modalities, such as T1, Flair, T1ce, and T2, each providing unique details about the tumor. Despite significant advancements in AI-based detection methods and the widespread use of MRI, building an accurate and efficient model for tumor classification remains a formidable task. To address these problems, the study proposes a novel approach that leverages deep residual learning with multi-modality image feature fusion. Unlike conventional methods that simply concatenate different modalities and input them directly into neural networks, our approach focuses on extracting and fusing distinct and complementary features from various MRI modalities. This strategy fully exploits the diverse information available, enhancing the performance of brain tumor image segmentation.

PROBLEM STATEMENT

- The Healthcare sector stands apart from other industries, prioritizing the highest level of care and services regardless of expenses.
- With the proven success of Deep Learning in various real-world applications, it now offers promising solutions with high accuracy for medical imaging, emerging as a pivotal method for future healthcare applications.
- Given the brain's central role in controlling bodily functions, automated brain tumor recognition in Magnetic Resonance Imaging (MRI) poses a formidable challenge due to the varied sizes and locations of tumors.
- However, Deep Learning algorithms present a viable approach to address this challenge, promising improved accuracy and a reduction in the required number of iterations.

STATISTICS

- World Brain Tumor Day - June 8.
- The World Health Organization (WHO) estimates that there are about 300,000 new cases of brain tumors diagnosed worldwide each year.
- It is estimated that there are over 2.5 million people living with a brain tumor globally.
- The median age at diagnosis for a primary brain tumor is 61 years
- In India, 40,000-50,000 people are diagnosed with a brain tumor every year, according to estimates.
- Out of these, about 20 percent are children. [[Source](#)]
- Brain cancer is estimated to be the 10th leading cause of cancer death in 2023 for both males and females in all age groups.

LITERATURE SURVEY

S.No	TITLE	ALGORITHM USED	ADVANTAGES	LIMITATIONS
1	Deep Learning and Optimized Learning Machine for Brain Tumor Classification	Pretrained CNN, SVM and KNN algorithms.	- Efficient feature extraction and fusion techniques. - Improved accuracy.	Resource intensive and time consuming approach.
2	Brain tumor segmentation based on optimized convolutional neural network and improved chimp optimization algorithm.	Optimized Convolutional Neural Network.	- Multi-Modality MRI Integration. - Improved Chimp Optimization Algorithm.	There is a lack of population diversity due to increased randomness.
3	DenseUNet+: A novel hybrid segmentation approach based on multi-modality images for brain tumor segmentation.	DenseUNet algorithm.	- Improved edge detection to prevent feature loss. - Improved performance with reduced training time.	Complexity increases as layers become more deep.

EXISTING METHODOLOGIES

EXISTING ALGORITHMS

- Convolutional Neural Networks (CNN)
- Deep Convolutional Generative Adversarial Network (DCGAN)
- Residual Neural Network (RESNET)
- CNN + UNET
- DNN + SVM

LIMITATIONS

- Decreased Data Efficiency.
- Limited Interpretability.
- Overfitting and Generalization.
- Computational Complexity.

OBJECTIVES

- To implement a system which results in classification and segmentation of brain tumours in MRI images.
- To utilize efficient and scalable algorithms to improve accuracy and efficiency of classification and segmentation.
- Explore and implement parallel processing techniques to leverage the power of multi-core CPUs or GPUs, enhancing the computational efficiency of both classification and segmentation tasks.
- Investigate the use of ensemble learning methods, such as combining multiple models or predictions, to enhance the robustness and accuracy of brain tumor classification and segmentation.

PROPOSED ARCHITECTURE

PROPOSED ALGORITHMS

▪ CLASSIFICATION

1. EfficientNetB3
2. Vision Transformer
3. EfficientNetV2
4. Convolutional Neural Networks

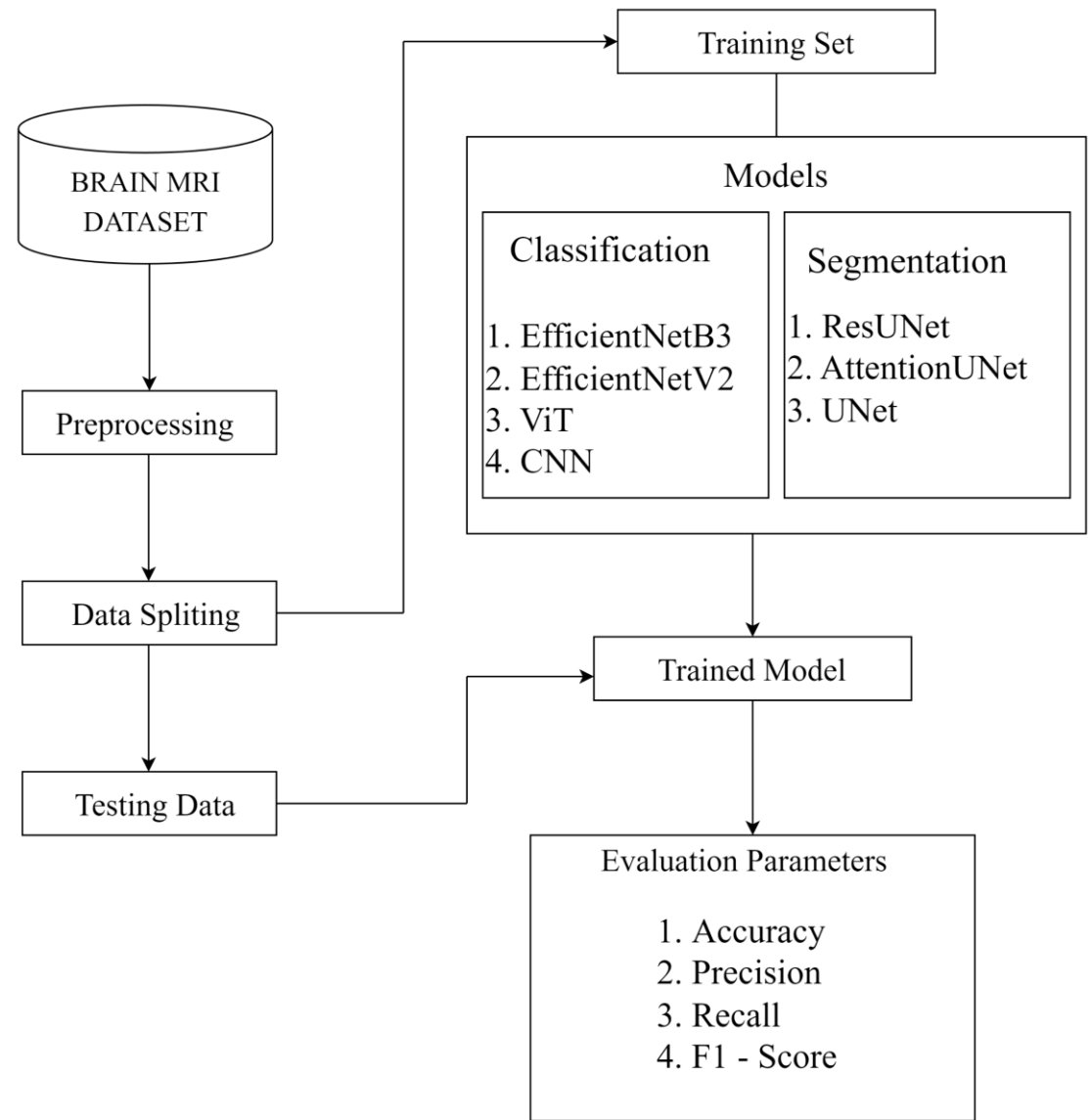
▪ SEGMENTATION

1. ResUNet
2. AttentionUNet
3. UNet

EFFICIENCY OF PROPOSED ALGORITHMS:

- Improved robustness.
- Flexible to different image resolutions.
- Spatial Localization

PROPOSED WORKFLOW



MODULES

- DATA COLLECTION
- DATA PRE-PROCESSING
- MODEL CREATION
- MODEL TRAINING
- MODEL EVALUATION
- DEPLOYMENT

TUMOR TYPES

Features	Glioma Tumor	Meningitis Tumor	Pituitary Tumor
Location	Central nervous system	Central nervous system	Pituitary gland
Cell of Origin	Glial cells	Inflammation of the meninges	Pituitary gland cells
Treatment	Surgery, radiation therapy, chemotherapy	Antibiotics, antiviral drugs, supportive care	Medications, surgery, radiation therapy
Symptoms	Headaches, seizures, neurological deficits	Headaches, fever, neck stiffness, photophobia, confusion	Vision problems, hormonal imbalances, headaches
Causes	Genetic factors, radiation exposure, unknown	Viral or bacterial infection	Often benign, genetic factors, hormonal imbalances
Age Group Affected	More common in adults	Children and Young Adults	Adults

EFFICIENTNETB3

- It is a part of the EfficientNet family, which is based on compound scaling that uniformly scales the network's depth, width, and resolution. EfficientNetB3 is deeper and wider compared to its smaller counterparts, offering a good balance between model size and performance.
- EfficientNetB3 incorporates advanced techniques such as squeeze-and-excitation blocks and swish activation functions, which help enhance its performance.
- The architecture of EfficientNetB3 allows it to achieve state-of-the-art results on benchmark datasets while being computationally efficient.

ADVANTAGES

- High performance.
- Robustness to overfitting.
- Transfer learning effectiveness.

EFFICIENTNETB3 ARCHITECTURE



The basic building block of the EfficientNet architecture is **Mobile inverted Bottleneck convolution (MBConv)** with squeeze and excitation optimization.

Block N°	Layer Name	Resolution	Filter Size	Number of Layers
1	Conv	300 × 300	3 × 3	1
2	MBConv1	150 × 150	3 × 3	2
3	MBConv6	150 × 150	5 × 5	3
4	MBConv6	75 × 75	3 × 3	3
5	MBConv6	38 × 38	3 × 3	5
6	MBConv6	19 × 19	5 × 5	5
7	MBConv6	10 × 10	5 × 5	6
8	MBConv6	10 × 10	3 × 3	2
9	Conv	10 × 10	1 × 1	1
10	Global pooling	10 × 10		1
11	Dense layer	10 × 10		1

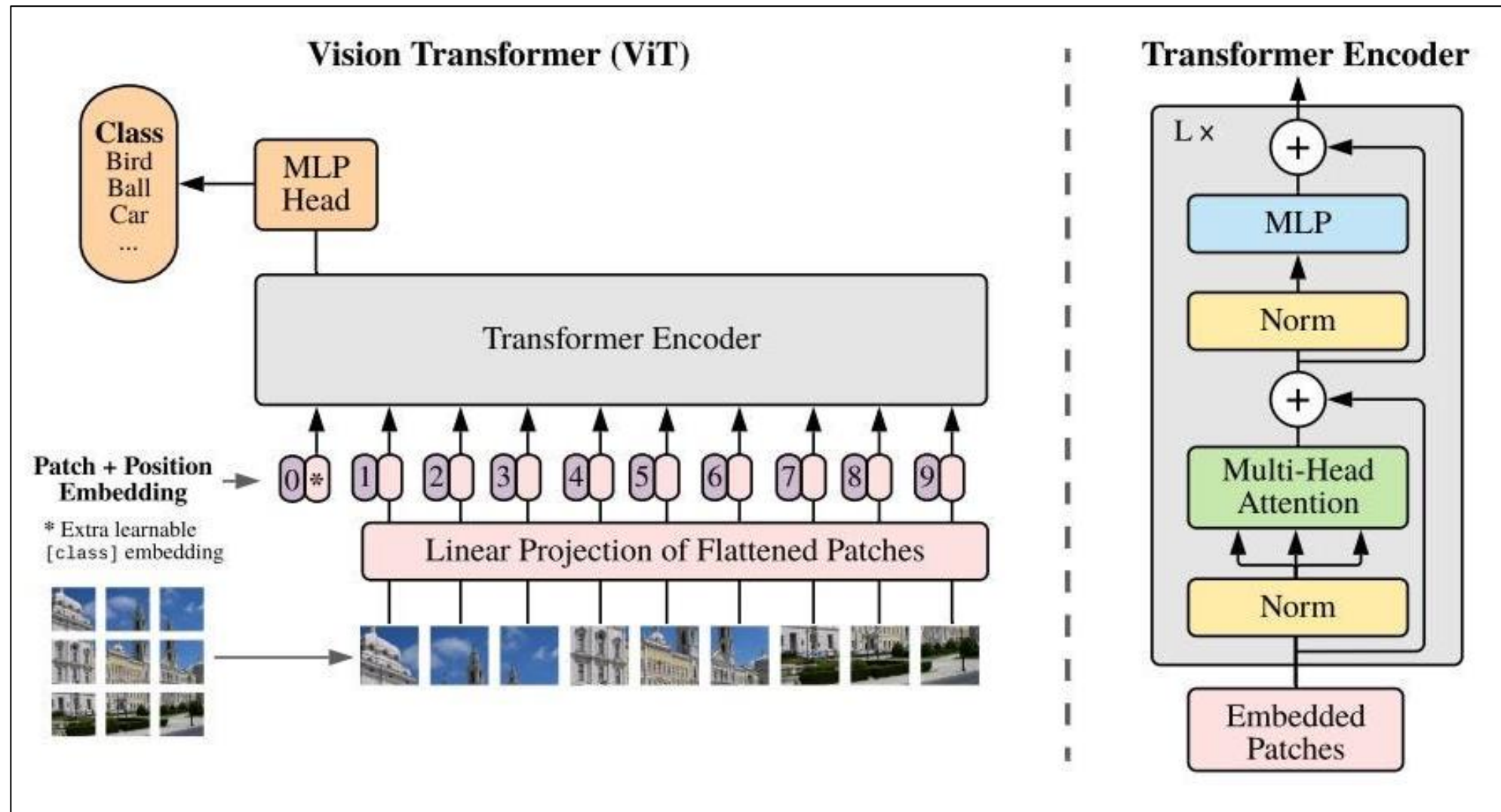
VISION TRANSFORMER

- Vision Transformer is a model for image classification that employs a Transformer-like architecture over patches of the image.
- An image is split into fixed-size patches, linearly embedded, position embeddings are added, and the resulting sequence of vectors is fed to a standard Transformer encoder.
- In order to perform classification, the standard approach of adding an extra learnable “classification token” to the sequence is used.
- Unlike Convolutional neural networks (CNNs), ViT can process images of varying resolutions by adjusting the number of patches and size of the patch embeddings.

ADVANTAGES

- Global Context Understanding.
- Capturing Long-Range Dependencies.

VISION TRANSFORMER ARCHITECTURE



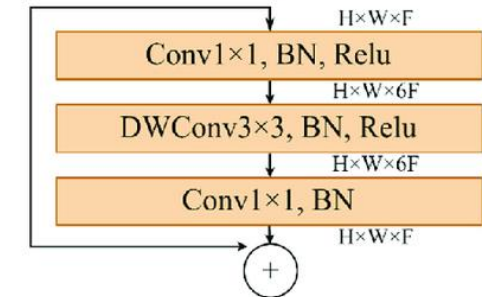
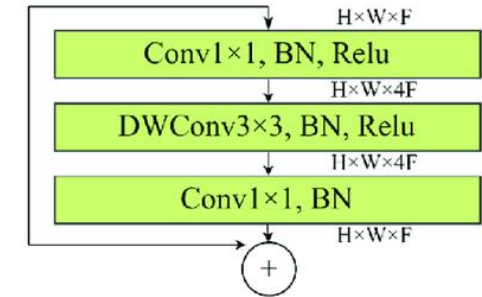
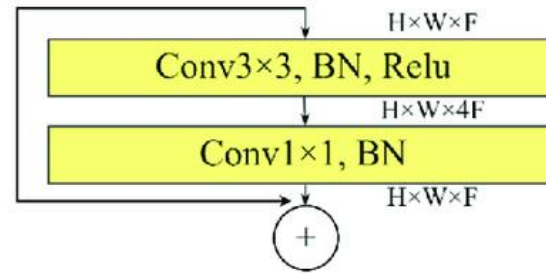
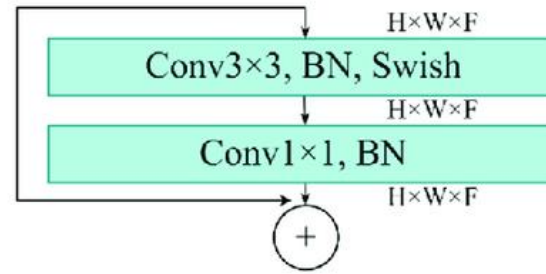
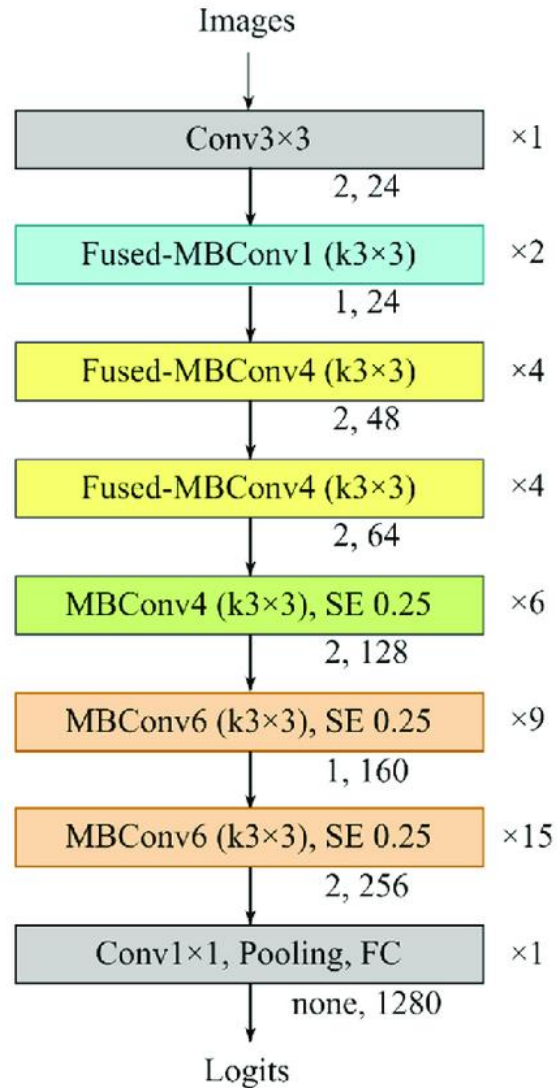
EFFICIENTNETV2

- EfficientNetV2 are a family of image classification models, which achieve better parameter efficiency and faster training speed than prior arts.
- Built upon EfficientNetV1, EfficientNetV2 models use Neural Architecture Search (NAS) to jointly optimize model size and training speed (faster training and inference speed).
- EfficientNetV2 models train much faster than state-of-the-art models (6.8x smaller).
- **Neural Architecture Search (NAS)** - automating the design of artificial neural networks to design networks that are on par or outperform hand-designed architectures.

ADVANTAGES

- Compound Model Scaling.
- Efficient channel attention mechanism.
- Neural Architecture Search.

EFFICIENTNETV2 ARCHITECTURE



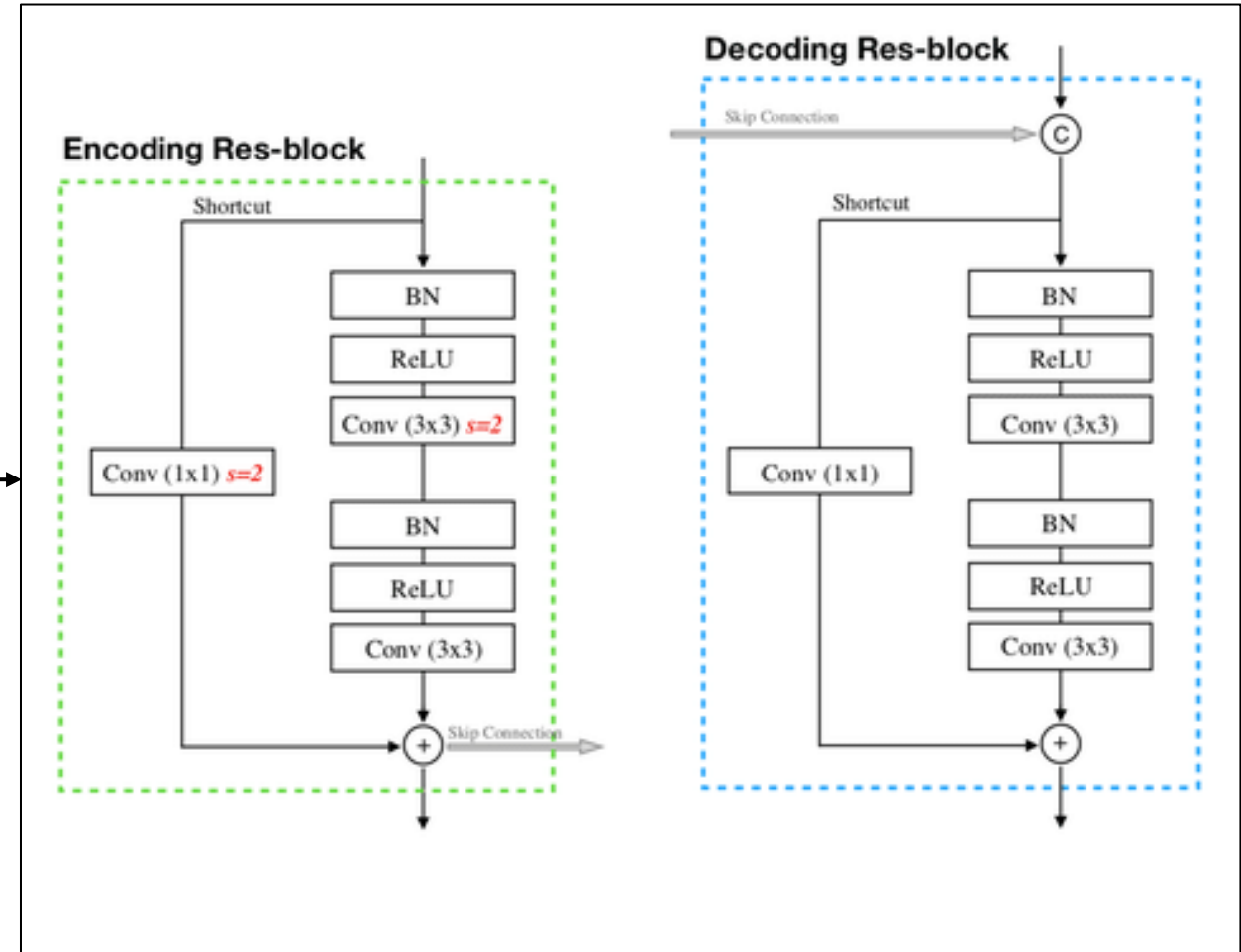
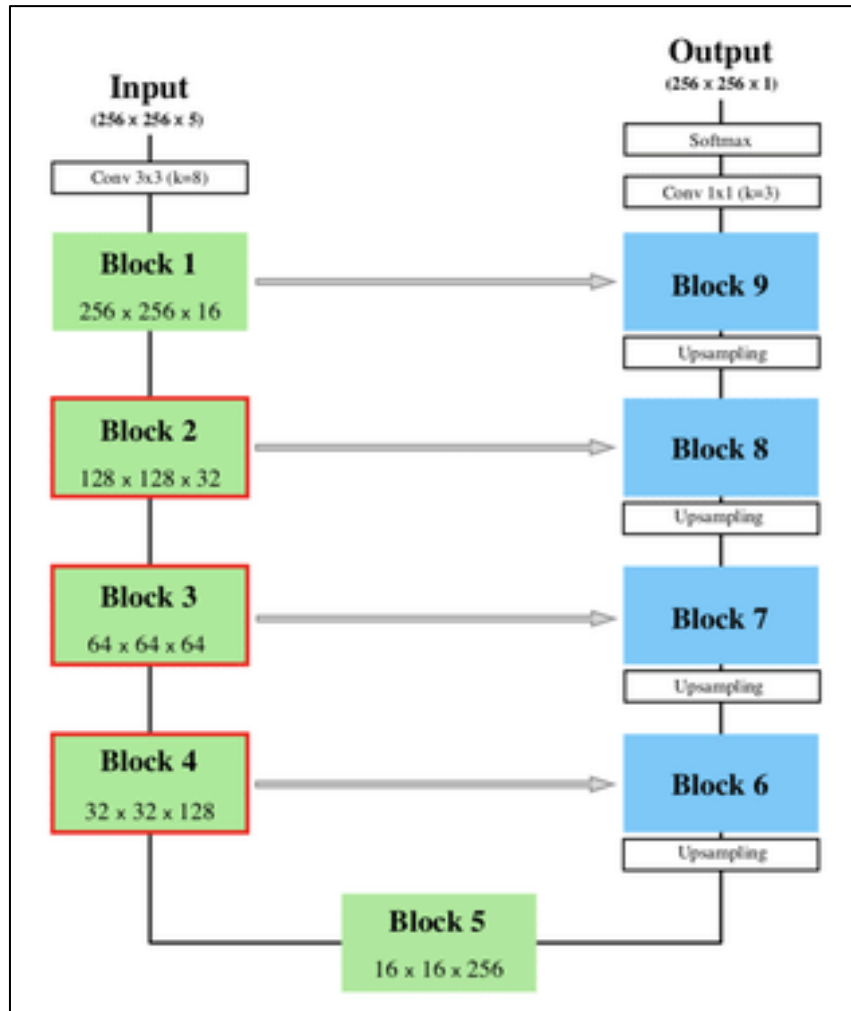
DEEP RESIDUAL UNET (RESUNET)

- RESUNET is a fully convolutional neural network that is designed to get high performance with fewer parameters. It is an improvement over the existing UNET architecture which help in training deeper networks without the vanishing gradient problem.
- The RESUNET consists of an encoding network (4 blocks), decoding network (4 blocks) and a bridge (1 block) connecting both these networks, just like a U-Net.
- The output of the last decoder passes sigmoid activation which gives the pixel-wise classification.
- RESUNET takes the advantage of both the UNET architecture and the Deep Residual Learning.

ADVANTAGES

- Skip Connections.
- Effective Semantic Segmentation.
- Handles high-resolution images.

RESUNET ARCHITECTURE



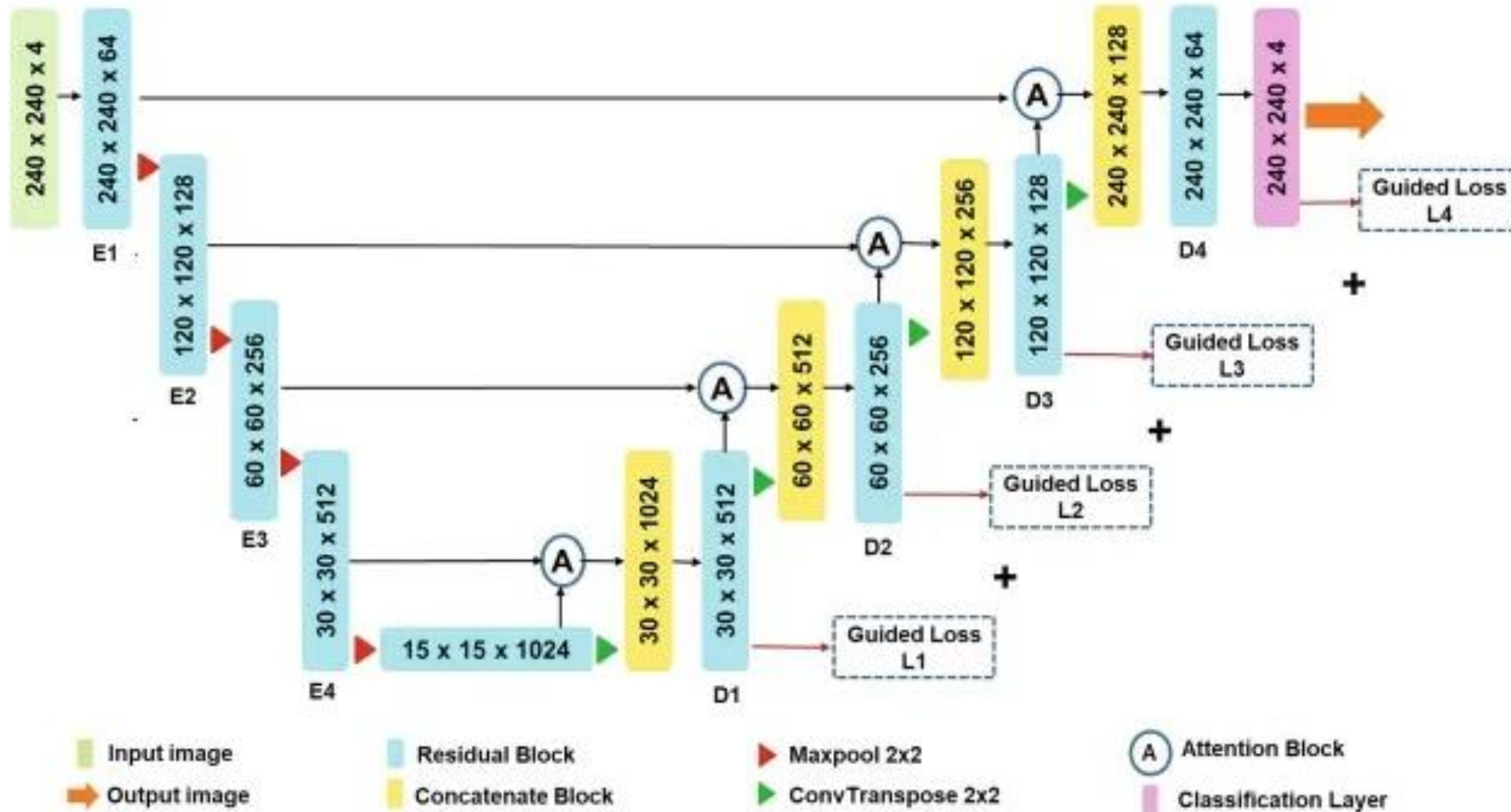
ATTENTION-UNET

- Attention U-Net integrates an attention mechanism into traditional U-Net architecture. This mechanism allows the network to focus on relevant parts of the input image, improving segmentation accuracy.
- Helps to selectively aggregate features from different levels of image based on their importance.
- These models can be trained efficiently and effectively using standard optimization techniques.
- The attention mechanism allows the network to adapt for different spatial locations, improving its ability to handle varying object sizes and shapes.

ADVANTAGES

- Effective Handling of Class Imbalance.
- Reduced Computational Complexity.
- Better Handling of Object Occlusion

ATTENTION-UNET ARCHITECTURE



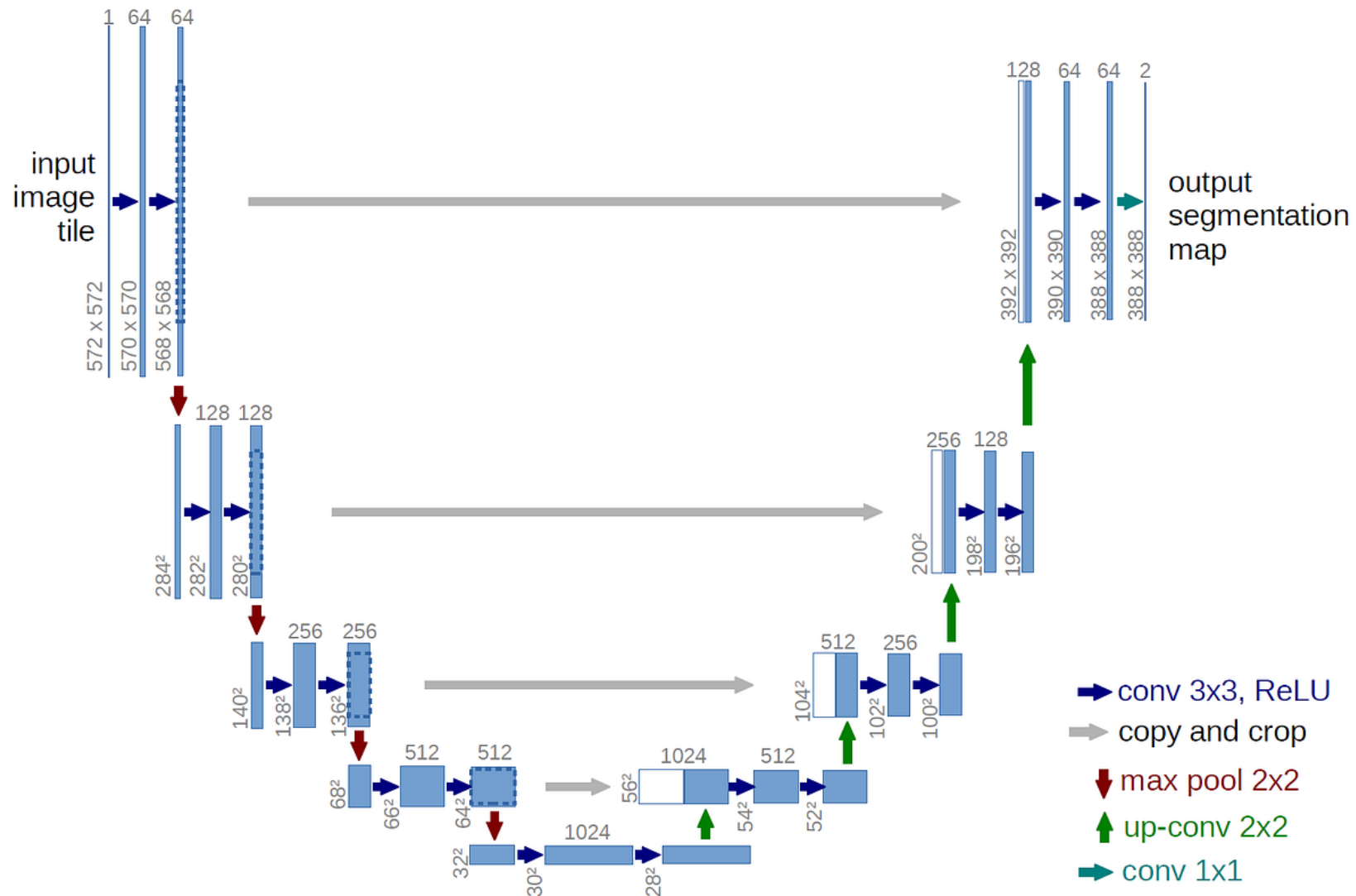
UNET

- U-Net is a convolutional neural network architecture designed for image segmentation tasks.
- It usually features four contracting paths to capture context and four expanding paths for precise localization.
- U-Net uses skip connections between mirrored layers to preserve spatial information. Each block contains several convolutional layers.
- U-Net is widely used in biomedical image segmentation and other domains due to its versatility.

ADVANTAGES

- Fast Inference
- Preservation of Spatial Information.
- Strong Performance.

UNET ARCHITECTURE



MODEL METRICS

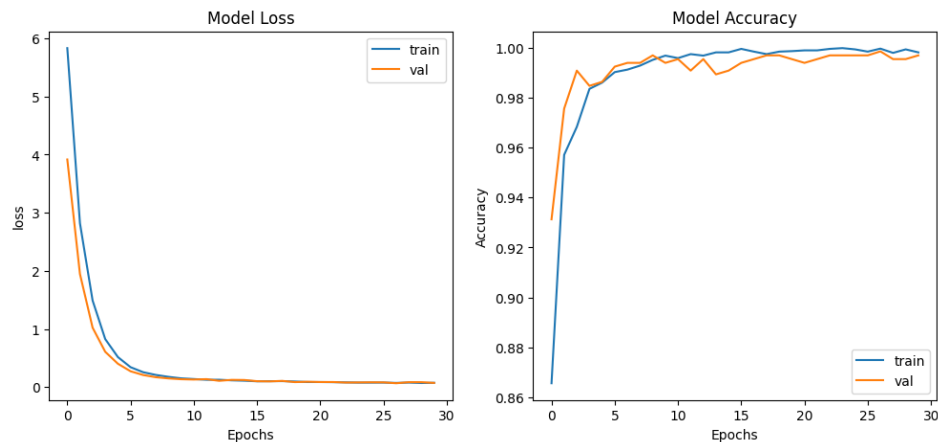
CLASSIFICATION

Model	Loss	Accuracy	Precision	Recall	F1-Score
EfficientNet-B3	0.0713	0.9993	0.99	0.99	0.99
EfficientNet-V2	0.4683	0.9527	0.95	0.95	0.95
ViT	0.4803	0.9509	0.94	0.93	0.93
CNN	0.1625	0.9568	0.94	0.94	0.94

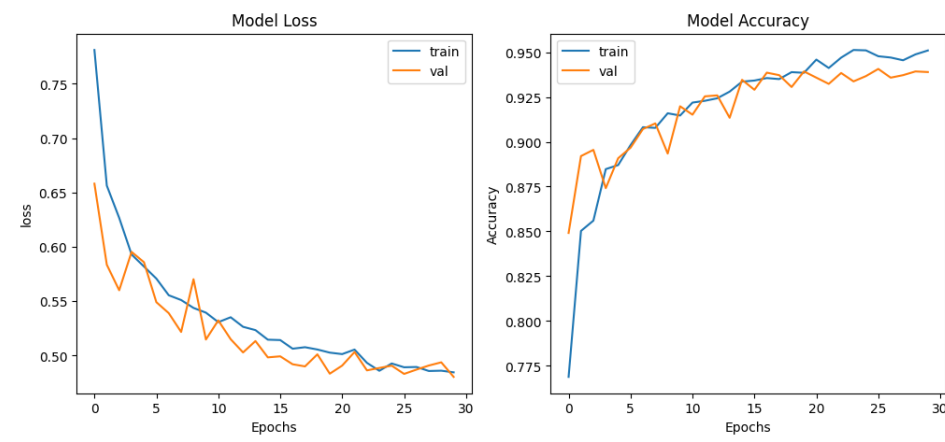
SEGMENTATION

Model	Focal Tversky Loss	Tversky Score	IOU Coefficient	Dice Coefficient
ResUnet	0.2007	0.8819	0.8243	0.8589
Attention Unet	0.2879	0.8056	0.6577	0.7901
Unet	0.2189	0.8217	0.7907	0.8590

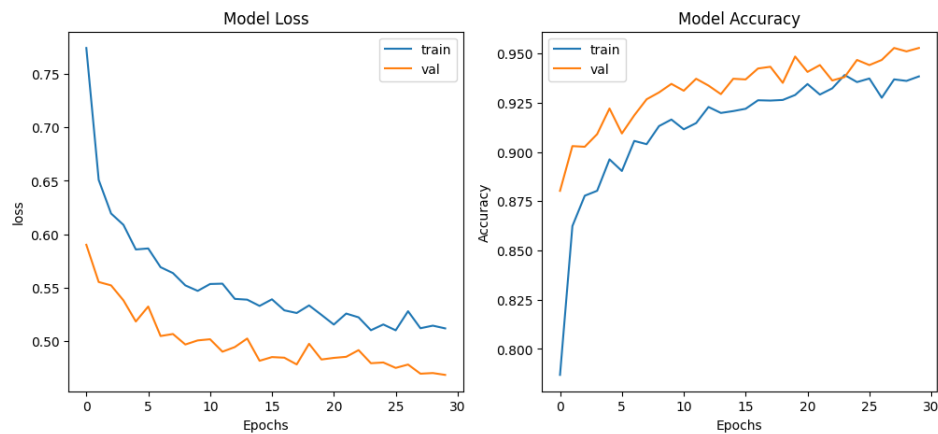
CLASSIFICATION RESULTS



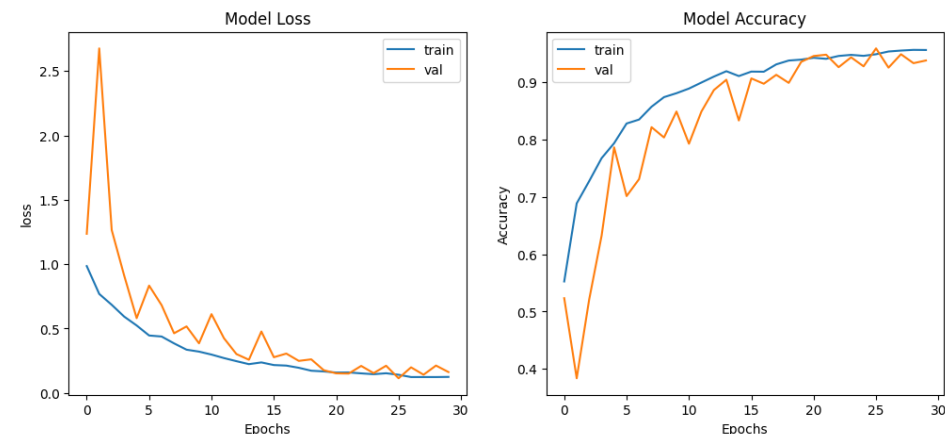
EfficientNet-B3



Vision Transformer



EfficientNet-V2



CNN

SEGMENTATION RESULTS

TVERSKY SCORE

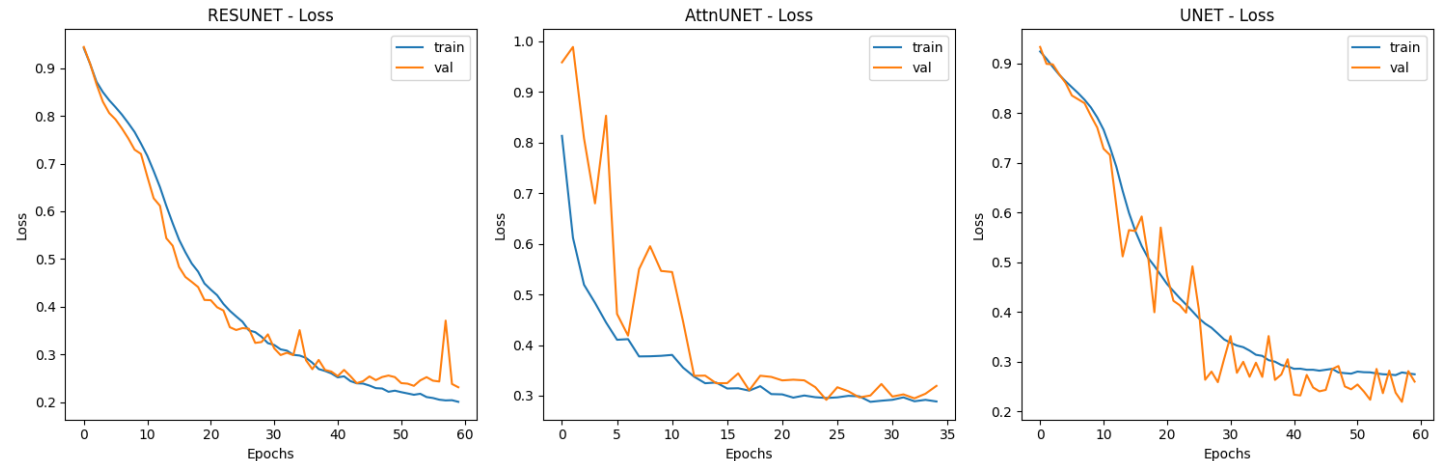
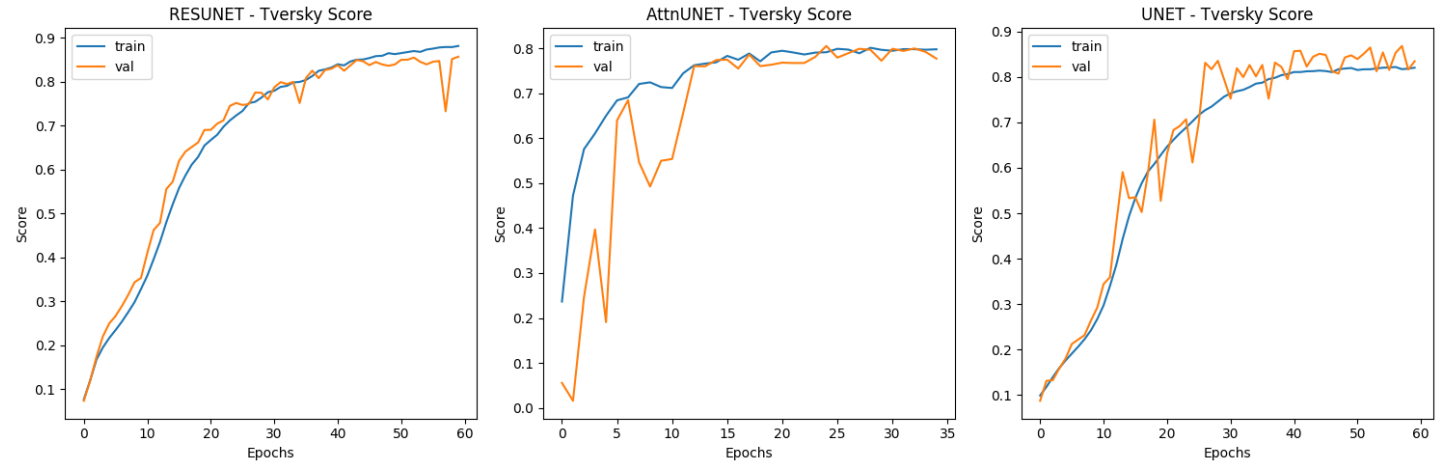
- extension of Jaccard index and F1 score.
- measure the dissimilarity between two sets, such as predicted and ground truth binary masks.

$$TI = \frac{TP}{TP + \alpha FN + \beta FP}$$

FOCAL TVERSKY LOSS

- generalisation of the tversky score.
- γ is a parameter that controls the non-linearity of the loss.

$$FTL = (1 - TI)^\gamma$$



SEGMENTATION RESULTS

IOU COEFFICIENT

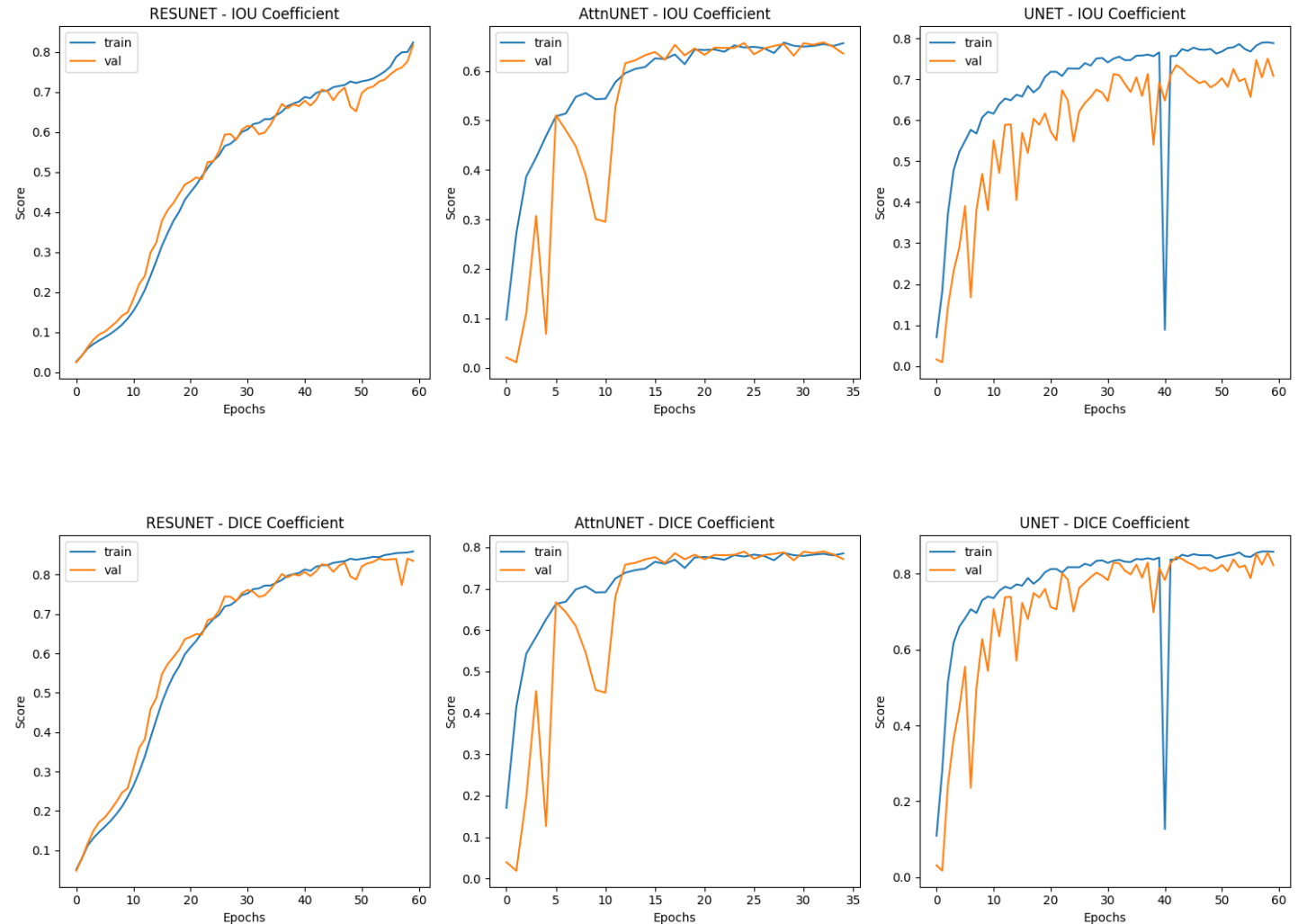
- calculates the ratio of the intersection of the actual and predicted masks to their union.
- IoU coefficient ranges from 0 to 1.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

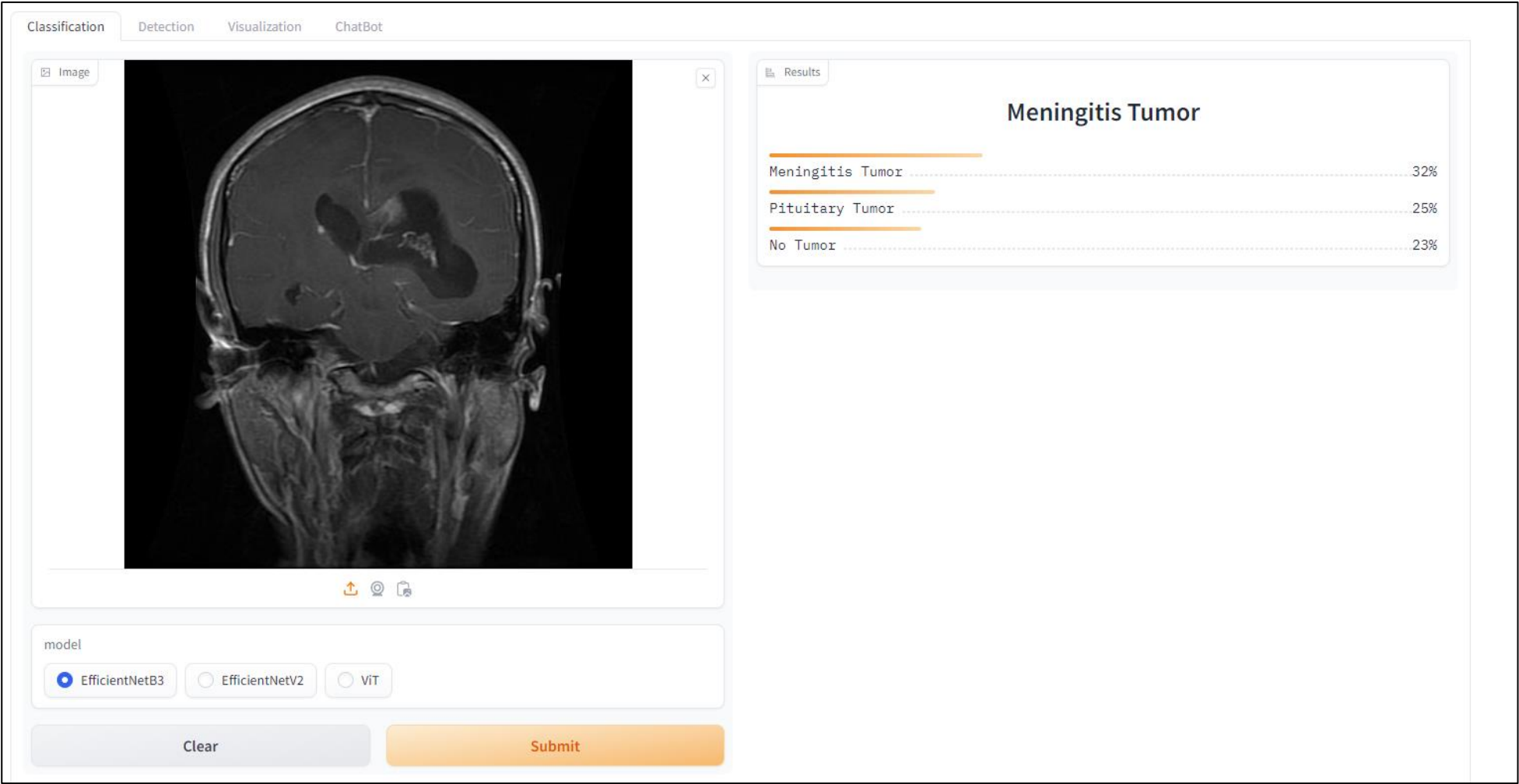
DICE COEFFICIENT

- quantify similarity between two sets.
- measuring the overlap between the predicted segmentation mask and the ground truth mask

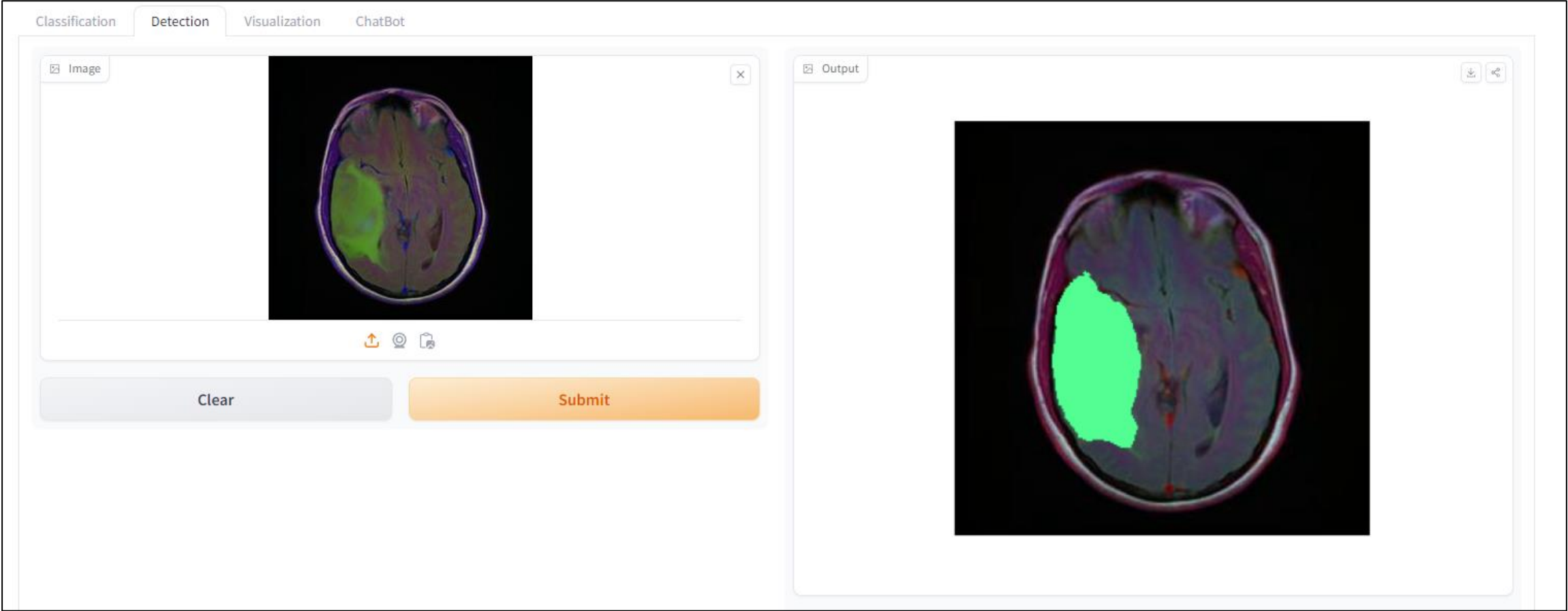
$$dice(A, B) = \frac{2|A \cap B|}{|A| + |B|}$$



SCREENSHOTS



SCREENSHOTS



SCREENSHOTS

Classification

Detection

Visualization

ChatBot

What are some of the treatment options for Brain Tumor?



The treatment options for brain tumors depend on various factors, including the type, size, and location of the tumor, as well as the patient's age, overall health, and personal preferences. Here are some common treatment options:

1. Surgery: If the tumor is located in a surgically accessible area, surgery may be recommended to remove as much of the tumor as possible. In some cases, surgery may be curative, especially for benign tumors.
2. Radiation therapy: Radiation therapy uses high-energy beams, such as X-rays or protons, to destroy tumor cells. It can be used alone or in combination with surgery.
3. Chemotherapy: Chemotherapy uses drugs to kill tumor cells. It can be given orally, intravenously, or through a wafer that is implanted in the brain during surgery.
4. Targeted therapy: Targeted therapy uses drugs that target specific genetic mutations or abnormalities in tumor cells. This type of therapy is still in the experimental stages for brain tumors, but shows promise for certain types of tumors.
5. Immunotherapy: Immunotherapy uses the body's





Retry

Undo

Clear

Type a message...

Submit

Examples

What is Brain Tumor and its types?

What is a tumor's grade? What does this mean?

What are some of the treatment options for Brain Tumor?

What causes brain tumors?

If I have a brain tumor, can I pass it on to my children?

CONCLUSION

- The application of various deep learning techniques and algorithms has significantly proved effective in the field of brain tumor classification and detection. These methods have proved to be more accurate and efficient in analyzing medical imaging data, providing valuable insights.
- EfficientNets, have proven effective in handling complex medical image datasets. Vision Transformers leverage attention mechanisms, and show promise in capturing intricate patterns within medical images, enhancing the interpretability of the classification process. Res-Unet excels in accurately delineating tumor boundaries, aiding in precise tumor localization.
- The proposed classification model (EfficientNet-B3) achieves an accuracy of 99%, whereas the proposed segmentation (Res-Unet) model achieves a loss of 0.2007 and a focal Tversky score of 0.8819 (88.19%).
- The future prospects of brain tumor classification and detection through deep learning are promising. Continued research and development may lead to even more sophisticated architectures, optimization techniques, and data augmentation methods, further improving model performance. Integration with multimodal data, such as combining imaging with clinical information, could enhance the overall diagnostic capabilities.

FUTURE ENHANCEMENTS

- Using Generative Adversarial Networks (GANs)
- Integration of Multi-Modal Imaging
- Utilizing GPU-accelerated computing.
- Large-Scale Datasets.
- Interactive and Real-Time Segmentation
- Integration with Clinical Data.
- Validation and Benchmarking.

REFERENCES

- [1] Atika Akter, Nazeela Nosheen, Sabbir Ahmed, Mariom Hossain, Mohammad Abu Yousuf, Mohammad Ali Abdullah Almoyad, Khondokar Fida Hasan, Mohammad Ali Moni,. (2023) “Robust clinical applicable CNN and U-Net based algorithm for MRI classification and segmentation for brain tumor”, Expert Systems With Applications 238, Article 122347.
- [2] Guying Zhang, Jia Zhou, Guanghua He, Hancan Zhu,. (2023) “Deep fusion of multi-modal features for brain tumor image segmentation”. Heliyon 9 (2023), Article e19266.
- [3] Halit Çetiner, Sedat Metlek,. (2023) “DenseUNet+: A novel hybrid segmentation approach based on multi-modality images for brain tumor segmentation”, Journal of King Saud University – Computer and Information Sciences 35 (2023), Article 101663.
- [4] Ramin Ranjbarzadeh, Payam Zarbakhsh, Annalina Caputo, Erfan Babae Tirkolae, Malika Bendeche, (2023) “Brain tumor segmentation based on optimized convolutional neural network and improved chimp optimization algorithm”, Computers in Biology and Medicine 168, Article 107723.

REFERENCES

- [5] Ronneberger, O., Fischer, P., Brox, T. (2015) “U-net: convolutional networks for biomedical image segmentation”, International Conference on Medical Image Computing and Computer-Assisted Intervention, 2015, pp. 234–241.
- [6] Sandhiya, B., Kanaga Suba Raja, S. (2023) “Deep Learning and Optimized Learning Machine for Brain Tumor Classification”, Biomedical Signal Processing and Control 89, Article 1057781746.
- [7] Wei Kwek Soh, Hing Yee Yuen, Jagath C. Rajapakse. (2023) “HUT: Hybrid UNet transformer for brain lesion and tumour segmentation”, Heliyon 9 (2023), Article e22412.
- [8] Zainon W.M., Alhassan A.M. (2020) “BAT Algorithm With fuzzy C-Ordered Means (BAFCOM) Clustering Segmentation and Enhanced Capsule Networks (ECN) for Brain Cancer MRI Images Classification”, IEEE Access 8 (2020) 201741–201751.
- [9] Zhao, X., Wu, Y., Song, G., Li, Z. , Zhang, Y., Fan, Y., (2018) “A deep learning model integrating FCNNs and CRFs for brain tumor segmentation”. Med. Image Anal. 43 (2018) 98–111.

THANK YOU